Name: SREEDEV P
Reg no: 25BYB0133

Code :

```
x <- c(10.5, 6.0, 8.7, 9.3, 11.8, 7.5, 15.0, 6.3, 8.5, 5.4)
y <- c(17.3, 14.0, 19.1, 14.5, 20.0, 16.3, 23.8, 14.0, 17.3, 13.3)

r <- cor(x, y)
cat("Correlation Coefficient (r):", r, "\n\n")

model_y_on_x <- lm(y ~ x)
a_yx <- coef(model_y_on_x)[1]
b_yx <- coef(model_y_on_x)[2]
cat("--- (i) Regression Equation of Y on X (Sales on Advertising) ---\n")
cat(sprintf("Equation: Y = %.4f + %.4f * X\n\n", a_yx, b_yx))

x_given <- 10
estimated_sales <- a_yx + b_yx * x_given
cat("--- (ii) Point Estimate for Rs 1000 advertising ---\n")
cat("Using X = 10 in the equation:\n")
cat(sprintf("Estimated Sales ('000 Rs): %.4f\n", estimated_sales))
cat(sprintf("In actual Rupees, this is Rs: %.2f\n\n", estimated_sales * 1000))

model_x_on_y <- lm(x ~ y)
a_xy <- coef(model_x_on_y)[1]
b_xy <- coef(model_x_on_y)[2]
cat("--- Regression Equation of X on Y (Advertising on Sales) ---\n")
cat(sprintf("Equation: X = %.4f + %.4f * Y\n\n", a_xy, b_xy))

cat("--- Regression Coefficients ---\n")
cat(sprintf("b_yx (Slope of Y on X): %.4f\n", b_yx))
cat(sprintf("b_xy (Slope of X on Y): %.4f\n\n", b_xy))

plot(x, y,
    main = "Regression of Sales on Advertising",
    xlab = "Level of Advertising ('00 Rs) [X]",
    ylab = "Level of Sales ('000 Rs) [Y]",
    pch = 19,
    col = "blue",
    xlim = c(4, 16),
    ylim = c(12, 25))

abline(model_y_on_x, col = "red", lwd = 2)

grid()
```

OUTPUT:

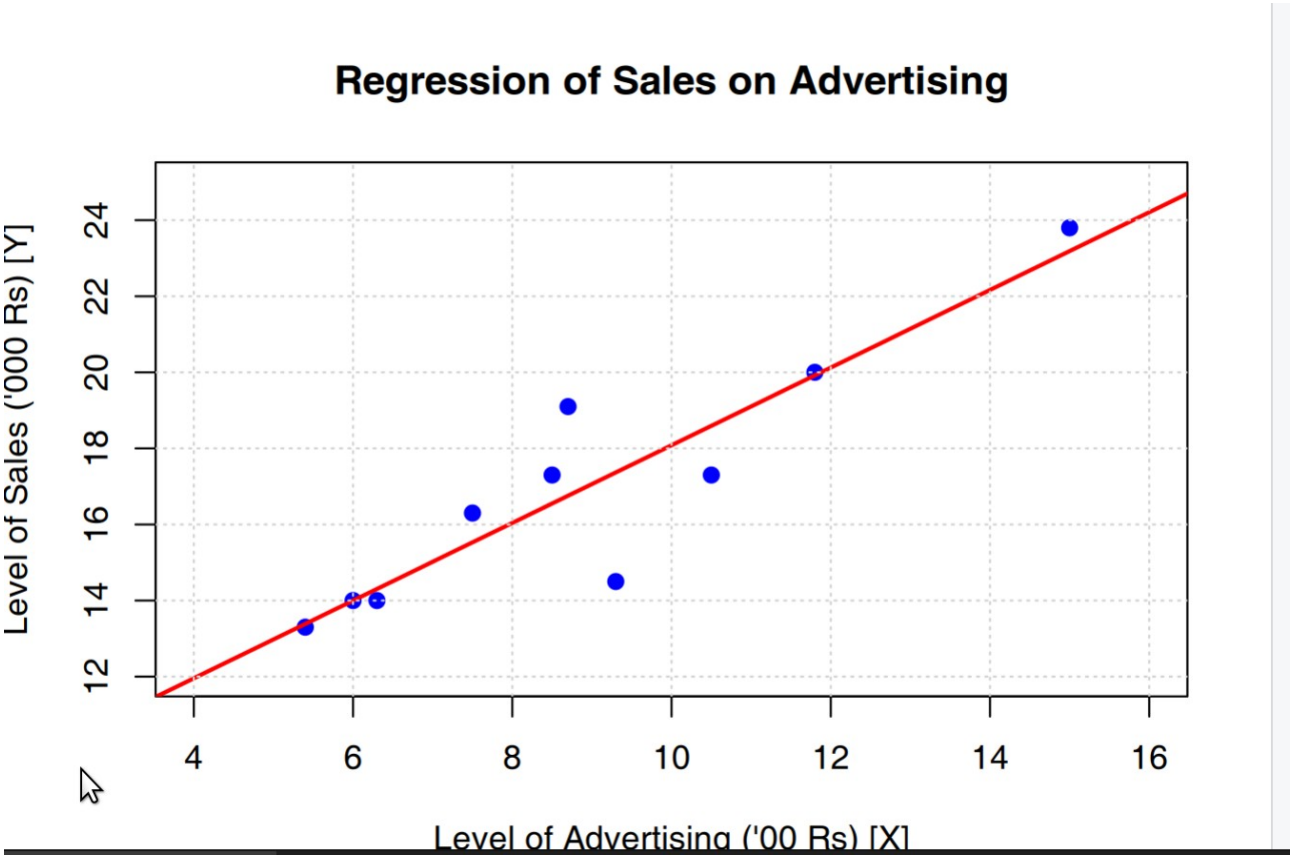
```
> r <- cor(x, y)
> cat("Correlation Coefficient (r):", r, "\n\n")
Correlation Coefficient (r): 0.9092068
```

```
--- (i) Regression Equation of Y on X (Sales on Advertising) ---
> cat(sprintf("Equation: Y = %.4f + %.4f * X\n\n", a_yx, b_yx))
Equation: Y = 7.8738 + 1.0209 * X
```

```
--- (ii) Point Estimate for Rs 1000 advertising ---
> cat("Using X = 10 in the equation:\n")
Using X = 10 in the equation:
> cat(sprintf("Estimated Sales ('000 Rs): %.4f\n", estimated_sales))
Estimated Sales ('000 Rs): 18.0830
> cat(sprintf("In actual Rupees, this is Rs: %.2f\n\n", estimated_sales * 100
0))
In actual Rupees, this is Rs: 18083.01
```

```
--- Regression Equation of X on Y (Advertising on Sales) ---
> cat(sprintf("Equation: X = %.4f + %.4f * Y\n\n", a_xy, b_xy))
Equation: X = -4.8328 + 0.8097 * Y

>
> cat("--- Regression Coefficients ---\n")
--- Regression Coefficients ---
> cat(sprintf("b_yx (Slope of Y on X): %.4f\n", b_yx))
b_yx (Slope of Y on X): 1.0209
> cat(sprintf("b_xy (Slope of X on Y): %.4f\n\n", b_xy))
b_xy (Slope of X on Y): 0.8097
```


Regression of Sales on Advertising
Level of Sales ('000 Rs) [Y]
12
14
16
18
20
22
24
4
6
8
10
12
14
16
Level of Advertising ('00 Rs) [X]